COURSE DESCRIPTION			
Program	**Diploma in Engineering and Technology / Commercial Practice / Management**		
Course Title	**Environmental Sustainability and Ethics**		
Course Code	**2005**	**Semester**	**II**
Type of Course	**Theory**	**Contact Hours**	**45**
Teaching Scheme (L: T:P:R)	**3:0:0:0**	**Credits**	**0**
CIE Marks	**40**	**ESE Marks**	**60**

Introduction

Environmental Sustainability and Professional Ethics is a course designed for Diploma Engineering students that focuses on protecting the environment and developing ethical values in professional life. It helps students to understand the importance of conserving natural resources, reducing pollution, and promoting sustainable development for present and future generations.

Course Objectives

1	To impart essential knowledge about environment, ecosystem, pollution and legal awareness on environmental policy and laws.
2	To analyze the concept of ethics and develop ethical approach to all professions specifically in approach to environment.
3	To develop an interdisciplinary approach towards sustainable development and environmental management.
4	To provide an awareness on zero waste concepts and practices for implementing sustainable environmental management.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic knowledge in environmental science and ethics			Secondary Education

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Explain the ecosystem, environmental pollution, important international conventions reg. pollution and legal awareness on environmental policy and laws.	Understand
CO2	Analyze the significance of environmental ethics and create consciousness about extended ethical approaches to the environment.	Understand
CO3	Explain sustainable energy resources, their explorations and design a plan for conservation and sustainability in all development programmes.	Understand
CO4	Explain sustainable engineering principles, zero waste concepts and practices for environmental management.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	3	3	-	-	-	-
CO2	-	-	2	-	-	3	3	-	-	-	-
CO3	-	-	2	-	-	3	-	-	-	-	-
CO4	-	-	3	-	-	3	2	-	-	-	-
Total	-	-	7	-	-	12	8	-	-	-	-
Correlation Level	-	-	2.33	-	-	3	2.67	-	-	-	-
Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"											

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	

3	0	0	0	4.5	45	0	40	60	100	0	0	0	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus – Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	T
Module I	Ecosystem & Environment- Fundamental concepts	Concept and scope of environmental Science, Environment – definition, segments, Ecosystem - definition, components of ecosystem, energy flow in an ecosystem- food chain & food web. Environmental pollution: Concepts and definition- Pollutant, Classification of pollutants-Global, regional, local, persistent and non-persistent pollutants. Environmental issues related to ozone depletion, global warming, Case study- Bhopal gas tragedy, Minamata disaster, Chernobyl nuclear disaster, Gulf war oil spill. Important International Conventions / Conference a) Stockholm Conference – 1973. b) Basel Convention c) Montreal protocol – 1987 d) Kyoto protocol – 1997 e)Paris Agreement (2015) f) Earth summit g) Agenda 21 h) Copenhagen summit i)MDG and SDG (basic understanding only). Major environmental policy and laws of Government of India for the protection of air, water, soil, forest etc.a) Indian forest act – 1927 b) Air pollution control act 1981 c) The Water (Prevention and control of pollution) Act – 1974 d) The Environment (Protection) Act (EPA) 1986	12	0
Module II	Ethics- Professional Ethics- Environmental ethics	Ethics - term, etymology, nature and scope - types of ethics; normative Ethics, meta Ethics, applied ethics. ethical theories; deontology, consequentialism, virtue ethics- (definitions only). Personal and professional ethics; moral stand points of personal ethics and professional ethics- (definitions only). Types of professional ethics: business ethics, legal ethics, medical ethics, environmental ethics (definitions only). Importance and historical development of Environmental Ethics, ethical approaches to environment; anthropocentrism, biocentrism, ecocentrism.	10	0
Module III	Sustainable Development	Man – Environment relationship, anthropogenic effects on the natural environment (pollution, deforestation, soil erosion, mass extinction etc.). Concept of sustainability, need of sustainability, sustainable development – approaches and strategies. Environment friendly products and technologies – products and technologies designed to reduce environmental impact, conserve resources and promote sustainability. Equity in resource distribution and consumption. Overview of renewable energy sources (solar, wind, hydro, biomass), sustainable technologies in energy production and consumption (solar power, wind energy, lithium batteries etc.).-brief introduction only. Generalized approach to impact analysis, environmental audit, sustainable development initiatives in India (national solar mission, green India mission, The national river conservation plan, smart city projects, Swachh Bharat Abhiyan, The national clean energy fund, The national water mission etc.) (basic understanding only), emerging trends and future directions in environmental ethics and sustainability, Urbanization and its environmental impact, sustainable urban planning and green infrastructure (urban parks, green buildings, sustainable transportation etc.).	13	0
Module IV	Ethical engineering and environmental management	Sustainable Engineering Principles- definition, scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives.Energy recovery from waste: biochemical methods (landfill, composting etc) and thermochemical methods (incineration, gasification and pyrolysis) Concept of Carbon Credit, Carbon Footprint. Environmental management in the fabrication industry. ISO14000: Implementation in industries, Benefits. Biotechnology and environmental protection-Green chemistry for sustainable development- basic principle	10	0

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Environment and Ecosystem	Define environment and ecosystem and explain the segments of the environment.	L	CO1	Understand	1

Environment and Ecosystem	Discuss the components- biotic and abiotic components of ecosystem	L	CO1	Understand	1
Environment and Ecosystem	Explain the food chain and food web with two examples.	L	CO1	Understand	1
Environmental pollution	Define environmental pollution and pollutants.	L	CO1	Remember	1
Environmental pollution	Classify the pollutants and briefly discuss global, regional, local, persistent and non -persistent pollutants.	L	CO1	Understand	1
Environmental pollution	Discuss the environmental issues related to ozone layer depletion and global warming.	L	CO1	Understand	1
Environmental pollution	Write a review of the Bhopal gas tragedy and Minamata disaster.	L	CO1	Understand	1
Environmental pollution	Briefly discuss the Chernobyl nuclear disaster and Gulf war oil spill.	L	CO1	Understand	1
Important international conventions/ Conferences reg. environmental issues	Give an overview of Stockholm Conference – 1973, Basel Convention, Montreal protocol – 1987, Kyoto protocol – 1997	L	CO1	Understand	1
Important international conventions/ Conferences reg. environmental issues	Outline the importance of Paris Agreement (2015), Earth summit, Agenda 21, Copenhagen summit, MDG and SDG	L	CO1	Understand	1
Environmental policy and laws of Govt. of India	Briefly discuss the Indian forest act – 1927 and Air pollution control act 1981	L	CO1	Understand	1
Environmental policy and laws of Govt. of India	Outline the main provisions of The Water Act – 1974 (Prevention and control of pollution) and The Environment (Protection) Act (EPA) 1986.	L	CO1	Understand	1
Module II					
Ethics: notion and scope	Define Ethics and describe the types of ethics. normative ethics, meta ethics, applied ethics. ethical theories; virtue ethics (definitions only).	L	CO2	Remember	1
Ethics: notion and scope	Make differences between deontology and consequentialism.	L	CO2	Understand	1
Ethics: notion and scope	Examine the notion of virtue ethics.	L	CO2	Understand	1
Division of ethics	Elaborate the moral stand points of personal ethics and professional ethics.	L	CO2	Understand	1
Division of ethics	Elucidate Professional ethics and personal ethics.	L	CO2	Understand	1
Professional ethics-types	Define business ethics and bring out the features of Legal ethics.	L	CO2	Understand	1
Professional ethics-types	Explain medical Ethics and environmental ethics.	L	CO2	Remember	1
Significance & different approaches to environmental ethics	Elucidate the historical developments of environmental ethics.	L	CO2	Understand	1
Significance & different approaches to environmental ethics	Elaborate the different ethical approaches to the environment and explain anthropocentrism.	L	CO2	Understand	1
Significance & different approaches to environmental ethics	Differentiate between bio-centrism and eco-centrism.	L	CO2	Understand	1
Module III					
Anthropogenic effects on the environment and concept of sustainability.	Discuss the relationship between man and environment.	L	CO3	Understand	1
Anthropogenic effects on the environment and concept of sustainability.	Explain the impacts of human interference on the environment.	L	CO3	Understand	1
Anthropogenic effects on the environment and concept of sustainability.	Illustrate the need for sustainability	L	CO3	Understand	1
Renewable energy sources	Overview of renewable energy sources	L	CO3	Understand	1

Renewable energy sources	Discuss the sustainable technologies in energy production and consumption	L	CO3	Understand	1
Environmental impact analysis	Explain the basic objective of Environmental impact analysis	L	CO3	Understand	1
Environmental impact analysis	Discuss the key steps in the EIA process	L	CO3	Understand	1
Environmental impact analysis	Explain the purpose and benefits of Environmental audit	L	CO3	Understand	1
Emerging trends and future directions in environmental sustainability	Outline of sustainable development initiatives in India	L	CO3	Apply	1
Emerging trends and future directions in environmental sustainability	Describe the environmental impact of Urbanization	L	CO3	Understand	1
Emerging trends and future directions in environmental sustainability	Discuss the principles and strategies of sustainable urban planning	L	CO3	Understand	1
Emerging trends and future directions in environmental sustainability	Discuss the emerging trends in sustainability	L	CO3	Understand	1
Emerging trends and future directions in environmental sustainability	Explain the role of green infrastructure planning in achieving sustainable development	L	CO3	Apply	1
Module IV					
Sustainable Engineering Principles	Describe the basic concepts of sustainable engineering principles	L	CO4	Understand	1
Sustainable Engineering Principles	Explain triple bottom line based on social, economical and environmental sustainability.	L	CO4	Understand	0.5
Sustainable Engineering Principles	Explain the concept of sustainable development associated with economic growth and environmental conservation.	L	CO4	Understand	0.5
Sustainable Engineering Principles	Discuss the need for life cycle analysis.	L	CO4	Understand	1
Zero waste concept	Illustrate the successful zero waste initiatives with a case study	L	CO4	Apply	1
Zero waste concept	Explain 3R principle for the disposal of solid waste.	L	CO4	Understand	0.5
Zero waste concept	Explain the necessity for recycling of waste materials.	L	CO4	Understand	0.5
Zero waste concept	Explain various thermochemical energy recovery methods from solid waste.	L	CO4	Apply	1
Zero waste concept	Explain various biochemical energy recovery methods from solid waste.	L	CO4	Apply	1
Environmental Management	Define Carbon credit and Carbon footprint.	L	CO4	Understand	0.5
Environmental Management	Describe the importance of ISO 14000 series in the fabrication industry.	L	CO4	Understand	1.5
Biotechnology and Green Chemistry	Describe the basic principles of biotechnology.	L	CO4	Understand	0.5
Biotechnology and Green Chemistry	Define green chemistry	L	CO4	Understand	0.5
Biotechnology and Green Chemistry	Explain the twelve principles of green chemistry	L	CO4	Understand	1

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Present a case study on unsustainable energy projects that affect surrounding landscapes or ecosystems.	4.5
2	Write a report on local environment issues regarding air or water pollution.	4.5
3	Prepare a chart showing energy transfer in an ecosystem.	4.5

4	Analyze about environmental protection laws in India and submit a report on it.	4.5
	Module II	
5	Write a note on ethical theories based on specific relevant cases.	4.5
6	Create diagrams showing connections between ethical principles and environmental practices.	4.5
7	Make posters expressing their ethical stance on environmental issues.	4.5
8	Describe the key ethical approaches to the environment.	4.5
	Module III	
9	Submit a report on successful implementation of any renewable energy sources in India.	4.5
10	Prepare a waste audit report of the campus including the suggestions for zero waste approach.	4.5
11	Write the life cycle analysis report of a common product used in Kerala (eg: a coconut, bamboo or rubber - based product) and present findings on its sustainability.	4.5
12	Analyse the energy consumption pattern of the campus and propose sustainable alternatives to reduce consumption.	4.5
	Module IV	
13	Undertake a case study on plastic waste management in Kerala and prepare a report.	4.5
14	Write a report on different types of energy recovery methods.	4.5
15	Write a note on the applications of green chemistry.	4.5
	Total Self-Learning hours	67.5

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Environmental Sustainability and Professional Ethics		SITTTR Kalamassery

REFERENCE			
Sl. No.	Title	Author	Publication
1	Environmental Engineering Science	Nazaroff, William, Cohen, Lisa	Willy, New York, 2000, ISBN 10: 0471144940.
2	Professional Ethics and Human Values	R. S. Naagarazam	New Age International (p) Limited New Delhi, 2026
3	Manual of Ethics	John Mackenzie	London University Tutorial Press, 1950
4	Sustainable Engineering principles and practice	Bhavik R. Bakshi,	Cambridge University Press, 2019.
5	Environmental Science: Toward a sustainable future	Richard T. Wright	Pearson, 2008.
6	Energy, Society and Environment, Technology for sustainable future	Elliot David	Rutledge, 2003.
7	Green Chemistry	V.K. Ahulwalia	Ane Books Pvt, ltd, New Delhi, 2006

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	www.eco-prayer.org	I & IV
2	www.teriin.org	I & IV
3	www.indiaenvironmentportal.org.in	I & IV
4	https://sdgs.un.org	I & IV
5	www.conserve-energy-future.com	IV
6	https://www.greeksorgreeks.org	III

Assessment Methodology – Theory

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE
		Self-learning activities (Module 1)	Self-learning activities (Module 2)	Self-learning activities (Module 3)	Self-learning activities (Module 4)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)
Duration						2 hours	2 hours	2.5 hours
Exam mark		15	15	15	15	50	50	60
Converted to						20	20	
Marks	5	15				20		60
Total	40							60
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		Total Marks
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60